# ECE 447

## Lesson 37
## Binary System Performance, Part 2



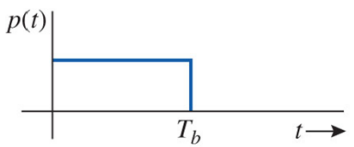
# Schedule and Admin

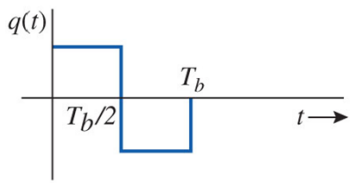
- Schedule.
  - Lesson 37 - Binary digital system performance (Part 2)
  - Lesson 38 - Error correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
  - **Final Exam** – 0730-1120, Saturday, 12 Dec
- Admin
  - **Lab 7**. Due tonight (Lesson 37, 23 Nov) to Gradescope.
  - **HW9**. Due Lesson 40 (Mon, 7 Dec) to Gradescope.
  - **Lab 8**. Previewed today; assigned Lesson 39 (Thu, 3 Dec), due Lesson 41 (Wed, 9 Dec).

# REVIEW

**Matched Filters and Binary Signaling Performance**

- Why?
- General binary case for real-valued pulses: $h(t) = p(T_b - t) - q(T_b - t)$
- $P_e = P_b = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$, where $E_{pq} = \int_0^{T_b} p(t)q(t)dt$ is the inner product or correlation of the two pulses
- Example problem: Find the matched filter impulse response $h(t)$ and calculate the total bit error probability $P_e$ assuming equiprobable transmission of 0 or 1

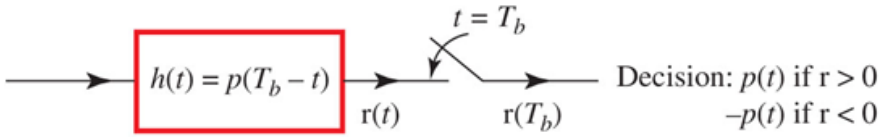


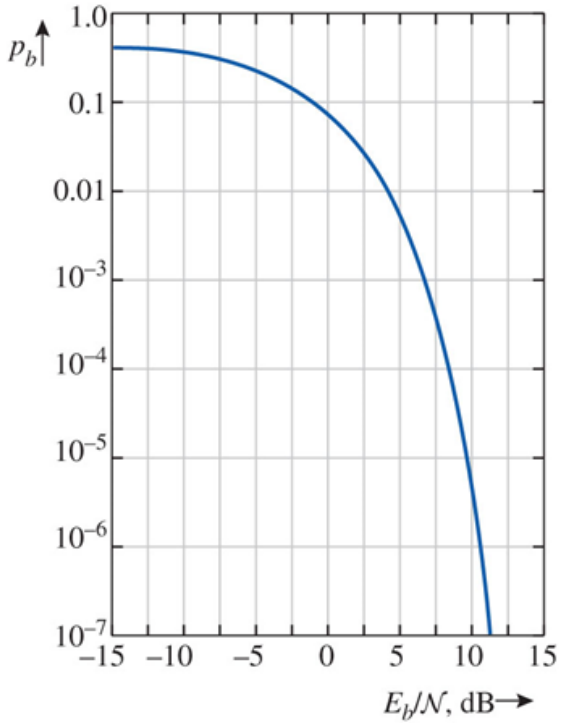
# Binary Signaling Performance (AWGN)

## Polar signaling, $q(t) = -p(t)$

- Use general equations for $P_e$ and $a_0$
- $E_p = E_q = \int_0^{T_b} |p(t)|^2 dt$
- $E_{pq} =$
- $P_e =$
- $a_0 =$
- $E_b = E_p \cdot P(m = 1) + E_q \cdot P(m = 0) =$
- BER depends on pulse energy not shape!




(a)

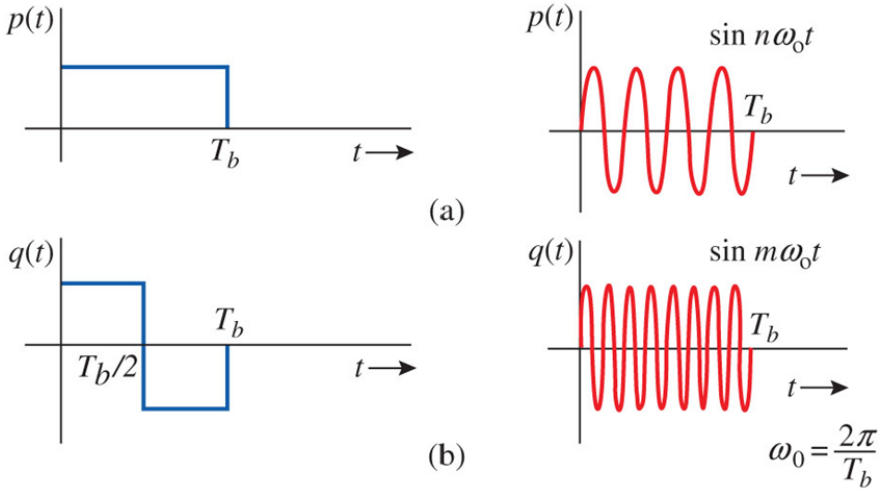



(b)

# Binary Signaling Performance (AWGN)

## **Orthogonal signaling**, $q(t)$ and $p(t)$ are orthogonal over $(0, T_b)$

- Use general equations for $P_e$ and $a_0$
- $E_q =$
- $E_{pq} =$
- $P_e =$
- $E_b = \frac{E_p + E_q}{2}$ assuming equiprobable
- $a_0 = \frac{1}{2}(E_p - E_q)$ (general form)

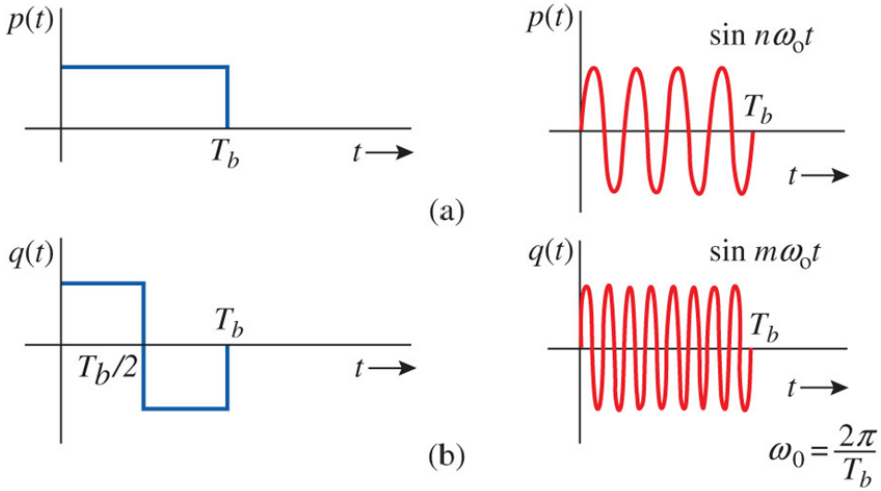



- Requires twice as much energy per bit (3db more power) to achieve same performance as polar signaling!

# Binary Signaling Performance (AWGN)

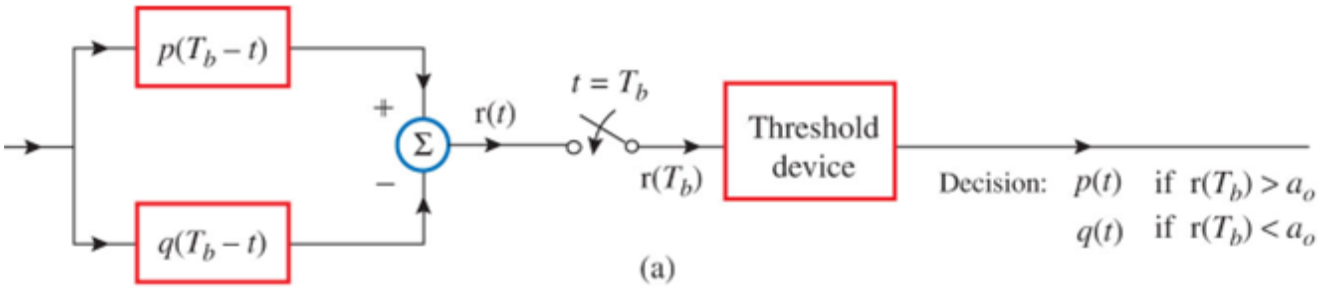
## **On-Off Signaling**, special case of orthogonal signaling

- Use general equations for $P_e$ and $a_0$
- $E_p = E_q$? Don't know, just leave $E_p$ and $E_q$
- $E_{pq} =$
- $P_e =$
- $E_b = \frac{E_p + E_q}{2} =$
- $a_0 = \frac{1}{2}(E_p - E_q) =$



# RECEIVERS

## Optimum binary threshold detector (baseband)




## Coherent detector for bandpass signals

